

T3 Weinberg-coefficient RGE report

$T3 - B$, $\alpha = -1$: $S_1 = (2, -\frac{1}{2})$, $S_2 = (2, \frac{1}{2})$, $F = (1, 0)$

Matched EFT RGE	Success	Flavor RGE	Success
Neutrino mass	Success	Numerical running	Not run
Observables	Not run	Internal benchmark	Success

1 Conventions and scope

Below the common heavy-particle threshold, the active theory is SMEFT.

$$t = \ln \mu,$$

$$V(H) = \frac{\lambda_H}{2} (H^\dagger H)^2.$$

The flavor construction assumes three lepton generations, a diagonal heavy-fermion mass basis, and common scalar masses.

2 Matched Weinberg coefficient

$$C_5 = -\frac{\hbar M_F^3 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\ + \frac{\hbar M_F^3 M_{S1}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\ - \frac{\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\ + \frac{\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)}$$

3 One-generation SMEFT check

The tensor calculation is required to reproduce the compact one-generation benchmark

$$16\pi^2 \beta_{C_5} = C_5 \left[-3g_2^2 + 2\lambda_H + 6|y_u|^2 + 6|y_d|^2 - |y_e|^2 \right]. \quad (1)$$

The RGE stage stops with an error if the calculated expression does not reduce to this benchmark.

4 Three-generation flavor result

The matched one-generation result is factorized as

$$C_5^{(1)} = F_{\text{loop}} \bar{y}_1 \bar{y}_2,$$

and lifted to the symmetric flavor matrix

$$(C_5)_{pq} = \frac{1}{2} \sum_r F_r \left[y_{1,pr}^* y_{2,qr}^* + y_{2,pr}^* y_{1,qr}^* \right]. \quad (2)$$

Writing this matrix as $K \equiv C_5$, the implemented RGE is

$$16\pi^2 \frac{dK}{d \ln \mu} = (2\lambda_H - 3g_2^2 + 2T)K - \frac{3}{2} [Y_e Y_e^\dagger K + K (Y_e Y_e^\dagger)^T], \quad (3)$$

$$T = \text{Tr} \left(Y_e Y_e^\dagger + 3Y_u Y_u^\dagger + 3Y_d Y_d^\dagger \right). \quad (4)$$

The fully expanded symbolic matrix is retained as a data file rather than printed here, because it is much less readable than the matrix equation above.

5 Neutrino-mass conversion

The current symbolic and numerical implementation uses

$$m_\nu = -v^2 C_5. \tag{5}$$

Normalization check required. Earlier function documentation stated $m_\nu = -(v^2/2)C_5$, whereas the executed code uses $m_\nu = -v^2 C_5$. The extracted operator normalization and this conversion must be checked together before final phenomenology.

6 Numerical running and observables

Numerical running was not requested. Supply `-numerical CONFIG.json` to include this section.

7 Saved data

Result	File
Matched coefficient	data/c5_coefficient.txt
Rendered matched coefficient	c5_coefficient.pdf
One-generation beta function	data/c5_beta.txt
Full flavor beta matrix	data/c5_flavor_beta_matrix.txt
Symbolic neutrino-mass matrix	data/neutrino_mass_matrix.txt